

calcium-data-provenance

Calcium Signalling Data Provenance

Summary

This document records what numerical data is needed for the platelet calcium signalling model, where each value comes from, and what discrepancies exist between sources.

Background: how calcium signalling works in platelets

Platelet activation produces a rapid rise in cytosolic calcium (Ca^{2+}), which triggers shape change, granule secretion, and aggregation. The calcium signal is generated by a tightly coupled set of pumps, channels, and sensors that span three compartments: the cytosol, the Dense Tubular System (DTS, an intracellular calcium store), and the extracellular space.

In summary: ADP binds P2Y1 $\rightarrow$ Gq activates PLC $\rightarrow$ PLC cleaves PIP2 to release IP3 $\rightarrow$ IP3 opens the IP3R channel in the DTS membrane, flooding Ca^{2+} into the cytosol $\rightarrow$ the Ca^{2+} rise drives activation. Three processes then restore resting state: SERCA pumps Ca^{2+} back into the DTS; PMCA ejects Ca^{2+} to the outside; and when the DTS empties, STIM1 senses this and gates Orai1 channels (SOCE) so that extracellular Ca^{2+} can restock the DTS.

Key term definitions

Term	Stands for	Role in model
Ca^{2+}	Calcium ion	The signalling molecule whose cytosolic concentration drives platelet activation
DTS	Dense Tubular System	Intracellular calcium store unique to platelets; equivalent to the endoplasmic/sarcoplasmic reticulum in other cells
IP3	Inositol 1,4,5-trisphosphate	Cytosolic second messenger produced by PLC ; binds to IP3R to open the DTS store
IP3R	IP3 Receptor	Ca^{2+} -permeable channel embedded in the DTS membrane; opens when IP3 binds; has complex 6-state gating including Ca^{2+} -dependent inactivation
SERCA	Sarco-Endoplasmic Reticulum Ca^{2+} -ATPase	ATP-powered pump that moves Ca^{2+} from cytosol back into the DTS; acts as the “refill” mechanism

Term	Stands for	Role in model
PMCA	Plasma Membrane Ca^{2+} -ATPase	ATP-powered pump on the outer membrane that ejects Ca^{2+} from the cell entirely; activated by calmodulin
STIM1	Stromal Interaction Molecule 1	Ca^{2+} sensor protein sitting in the DTS membrane; when DTS Ca^{2+} falls, STIM1 clusters and moves to trigger Orai1
Orai1	— (also called CRACM1)	Pore-forming subunit of the CRAC channel (Calcium Release-Activated Calcium channel) in the plasma membrane; gated by STIM1
SOCE	Store-Operated Ca^{2+} Entry	The overall process by which DTS depletion (sensed by STIM1) opens Orai1 channels and lets extracellular Ca^{2+} refill the cell
CRAC	Calcium Release-Activated Calcium	The ion channel formed by Orai1; opened by STIM1 binding; carries the SOCE current
PLC	Phospholipase C	Enzyme that cleaves PIP2. PLC isoforms are activated by GPCRs ($\text{ADP} \rightarrow \text{P2Y1} \rightarrow \text{Gq} \rightarrow \text{PLC}$). PLC isoforms by receptor tyrosine kinases ($\text{GPVI} \rightarrow \text{Syk} \rightarrow \text{PLC 2}$)
PIP2	Phosphatidylinositol 4,5-bisphosphate	Membrane phospholipid substrate for PLC; hydrolysed to yield IP3 (cytosolic) + DAG (membrane)
DAG	Diacylglycerol	Membrane lipid product of PIP2 hydrolysis; activates PKC; not directly modelled in calcium core
P2Y1	Purinergic receptor Y1	G-protein coupled receptor for ADP on platelet surface; activation leads to $\text{Gq} \rightarrow \text{PLC} \rightarrow \text{IP3}$ cascade
Gq / PLC	Heterotrimeric G protein Gq; Phospholipase C	Signal transduction bridge between P2Y1 and IP3 production
CaM	Calmodulin	Ubiquitous Ca^{2+} -binding protein; activates PMCA when Ca^{2+} is high; ~20,000 copies/platelet
MCU	Mitochondrial Calcium Uniporter	Channel that takes up Ca^{2+} into mitochondria when cytosolic Ca^{2+} is high; acts as buffer and integrator
PKG / PKA	Protein Kinase G / Protein Kinase A	Kinases activated by cGMP / cAMP; phosphorylate IP3R and other targets to inhibit activation (inhibitory pathway)

Source papers

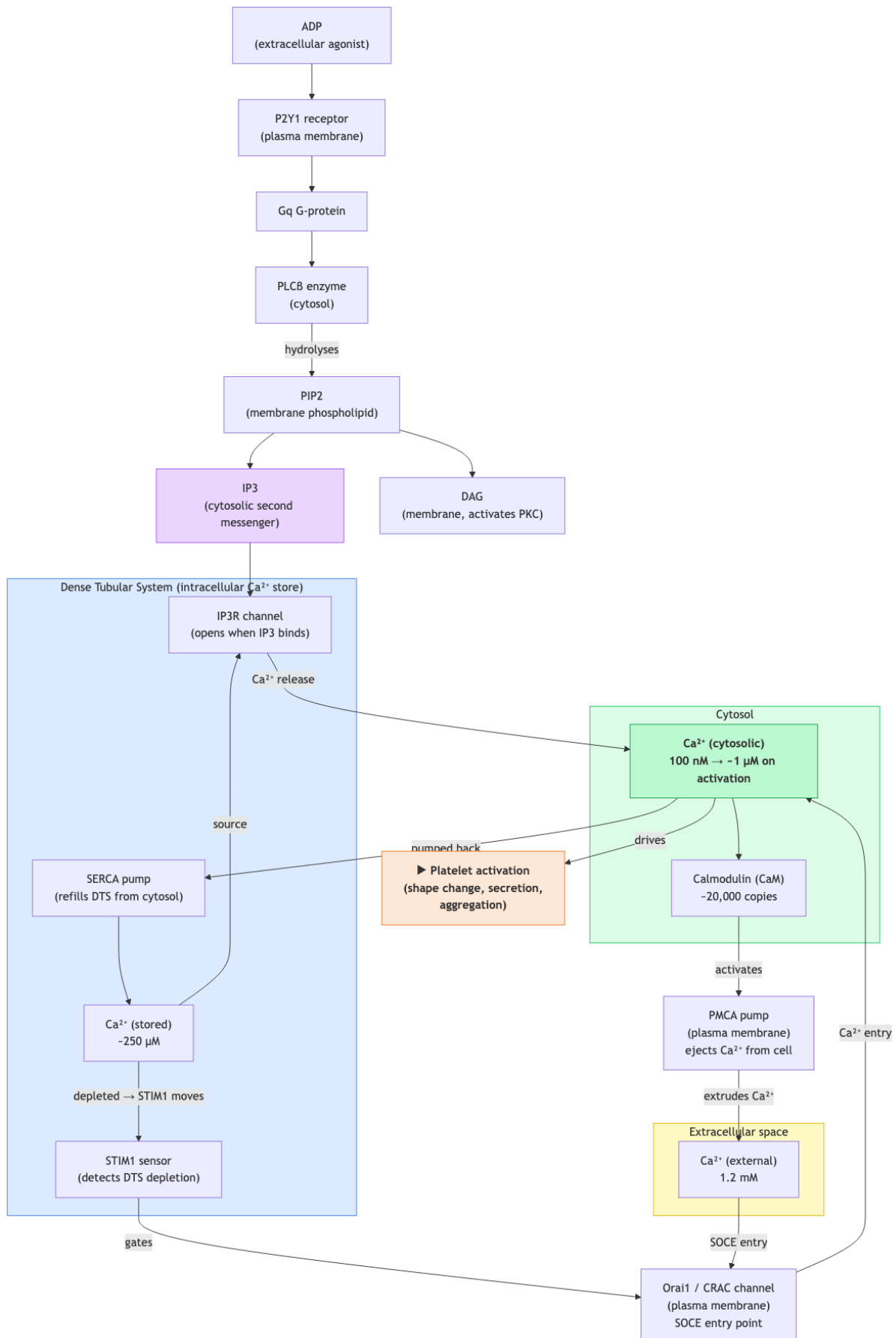


Figure 1: Calcium signalling flow diagram

Short ref	Full citation	Role
Purvis 2008	Purvis et al., Blood 112:4069–4079	Full upstream model: $\text{ADP} \rightarrow \text{P2Y1} \rightarrow \text{Gq} \rightarrow \text{PLC} \rightarrow \text{IP3} \rightarrow \text{Ca}^2$ 70 species, 77 reactions, 132 rate constants
Dolan 2014	Dolan & Diamond, Biophys J 106:2049–2060	Downstream Ca^2 model: IP3R + SERCA + PMCA + SOCE (STIM1/Orai1). 34 species, 35 reactions
Burkhart 2012	Burkhart et al., Blood 120:e73–e82	First comprehensive platelet proteome with copy numbers for ~3700 proteins
Sveshnikova 2015	Sveshnikova et al., Mol BioSyst 11:1052–1060	Compartmentalised Ca^2 model including mitochondria. Stochastic simulations
Sveshnikova 2025	Sveshnikova & Panteleev, Hämostaseologie 45(1)	Review/update: 29 species, 4 compartments, explicit parameter table
Ghatge 2026	Ghatge et al., J Thromb Haemost (in press)	MCU knockout data — mitochondrial uniporter role in Ca^2 and platelet function
Ajanel 2025	Ajanel et al.	MCU role in ITAM-dependent platelet activation
Shehwar 2025	Shehwar et al.	Review: platelets and mitochondria calcium connection
Kleppe 2018	Kleppe et al., J. Theor. Biol. 444:1-11	ODE model of inhibitory NO/cGMP/cAMP pathway ; includes PKA/PKG effects on IP3R
Dunster 2015	Dunster et al., PLoS Comput. Biol. 11:e1004118	ODE model of GPVI $\rightarrow$ Syk $\rightarrow$ PLC 2 pathway ; rate constants for upstream signalling
Mazet 2020	Mazet et al., Commun. Biol. 3:1-13	ODE model of the PI cycle ; rate constants for PIP2 hydrolysis and resynthesis

Compartment volumes

Compartment	Value	Source	Notes
Cytosol	6 fL	Purvis 2008 (ref 27)	“cytosolic volume of the platelet is approximately 6 fL”
Cytosol	3 fL	Sveshnikova 2025	Used in their model
DTS	2% of cytosol (median)	Purvis 2008	Monte Carlo estimation; range 0.5–5%
DTS	4.3% of cytosol	Purvis 2008	Direct measurement: glucose-6-phosphatase stain, DTS area 0.13 μm^2 vs cytosol 3.02 μm^2
DTS	1–10% of CYT	Dolan 2014 (refs 4,17,18)	Literature range
DTS	1.5 fL	Sveshnikova 2025	Used in their model; NB this is 50% of cytosol — much higher than other sources
Mitochondria	0.3 fL	Sveshnikova 2025	
Extracellular	infinite (fixed conc.)	All	Assumed well-mixed reservoir
Total platelet	<10 fL	Dolan 2014 (ref 9)	
Total platelet	~5 fL	Sveshnikova 2025	

Recommendation for the calcium core implementation: Use Purvis/Dolan consensus: cytosol ~6 fL, DTS ~2–4% of cytosol $\rightarrow$ ~0.12–0.25 fL. Flag the Sveshnikova 2025 DTS value (1.5 fL) as an outlier that needs explanation before adopting.

Resting state concentrations

Species	Value	Source	Notes
[Ca ²⁺]cyt	40–100 nM	Dolan 2014 (refs 9,36,44)	Consensus across literature
[Ca ²⁺]cyt	100 $\pm$ 10 nM	Purvis 2008	Used as homeostasis constraint
[Ca ²⁺]cyt	75 nM	Purvis 2008	Model resting value with basal ADP
[Ca ²⁺]cyt	13 nM (0.013 μM)	Sveshnikova 2025 Table 1	Lower than other sources
[Ca ²⁺]dts	100–400 μM	Dolan 2014	Literature range

Species	Value	Source	Notes
[Ca ²⁺] _{dt}	~250 μ M	Dolan 2014 (ref 45)	Measured with Fluo-5N
[Ca ²⁺] _{dt}	1000 μ M	Sveshnikova 2025 Table 1	Much higher; supports larger oscillations
[Ca ²⁺] _{ex}	1.2 mM	Dolan 2014	Used in their model
[Ca ²⁺] _{ex}	2.0 mM	Sveshnikova 2015/2025	
[IP3] resting	~200 nM (~1200 mol/platelet)	Purvis 2008 (ref 31)	Measured value
[IP3] resting	20–40 nM (<200 copies)	Dolan 2014	Model constraint from filtering
[IP3] resting	50 nM (0.05 μ M)	Sveshnikova 2025 Table 1	

Recommendation for the calcium core implementation: Start with [Ca²⁺]_{cyt} = 100 nM, [Ca²⁺]_{dt} = 250 μ M, [Ca²⁺]_{ex} = 1.2 mM, [IP3] = 50 nM. These are middle-of-road values consistent with Purvis and Dolan.

Protein copy numbers per platelet

Primary source: Burkhardt 2012 (proteomics). Secondary: Purvis 2008 (model estimation).

Protein	Copies/platelet	Source	Notes
IP3R type 1 (ITPR1)	2,400	Burkhardt 2012	
IP3R type 2 (ITPR2)	1,700	Burkhardt 2012	Most relevant: Sneyd/Purvis use type 2 kinetics
IP3R type 3 (ITPR3)	750	Burkhardt 2012	
IP3R total	~4,850	Burkhardt 2012	Sum of isoforms
IP3R (model estimate)	1,000–5,000	Purvis 2008	Mild/high-response configurations
SERCA2 (ATP2A2)	9,000	Burkhardt 2012	SERCA2b isoform
SERCA3 (ATP2A3)	16,300	Burkhardt 2012	SERCA3a isoform; dominant in platelets
SERCA total	~25,300	Burkhardt 2012	Much higher than IP3R; consistent with Purvis ratio
SERCA (model estimate)	~10	Purvis 2008	High-response configurations

Protein	Copies/platelet	Source	Notes
STIM1	7,400	Burkhart 2012	Ca ² sensor in DTS membrane
Orai1 (CRACM1)	1,700	Burkhart 2012	SOC pore-forming subunit (tetramer → ~425 channels)
PMCA4 (ATP2B4)	640	Burkhart 2012	Plasma membrane Ca ² -ATPase
ATPase 2C1	2,200	Burkhart 2012	Secretory pathway Ca ² ATPase
P2Y1	~150	Purvis 2008 (ref 42)	Low abundance GPCR
P2Y12	rank 296	Burkhart 2012	Copy number not explicitly given in text extract
PAR1	~1,000	Sveshnikova 2015/2025	Thrombin receptor
PLC 2	2,500	Burkhart 2012	
PLC 3	1,700	Burkhart 2012	
PLC 2	2,000	Burkhart 2012	
P2X1	1,400	Burkhart 2012	Ligand-gated Ca ² channel
Calmodulin	not extracted		Needed for PMCA activation
MCU	5,900	Burkhart 2012	Mitochondrial Ca ² uniporter
MICU1	1,400	Burkhart 2012	MCU regulator

Representative resting state — Dolan 2014 Table S1 (Supporting Material)

A single representative initial condition from the fully filtered population (satisfies all 4 homeostatic + dynamic constraints):

Protein (total copies/cell)	Sub-species	Copies
SERCA (11,892)	SERCA · E1	5,920
	SERCA · E2	5,927
	SERCA · E1 · Ca ²	6
	SERCA · E1P · Ca ²	7
	SERCA · E2P · Ca ²	4
	SERCA · E2P	28
IP3R (1,328)	IP3R · n (neutral)	809

Protein (total copies/cell)	Sub-species	Copies
PMCA (769)	IP3R · i1 (inhibited-1)	167
	IP3R · o (open)	261
	IP3R · a (active)	65
	IP3R · i2 (inhibited-2)	25
	IP3R · s (shut)	~0
	PMCA (free)	764
	Ca ²⁺ · PMCA	4
	PMCA(Ca ²⁺ CaM)	~0
	PMCA(Ca ²⁺ CaM) · Ca ²⁺	~0
	PMCA · CaM	~0
CaM (20,481)	CaM (free)	20,465
	Ca ²⁺ · CaM	15
	Ca ²⁺ · CaM	1
	(bound to PMCA)	~0
STIM (4,265 total)	STIM1 (free monomer)	438
	STIM1 (dimer, ×2)	11 dimers (=22 protein)
	STIM1 · Ca ²⁺ dts (DTS-bound)	3,805
Orai (1,447)	Orai (free)	1,477

Key observations from Table S1: - At rest, >99% of SERCA is shuttling between E1/E2 (unoccupied) - IP3R is mostly in neutral (61%) and open (20%) states; very little is “shut” - Almost all CaM is Ca²⁺-free (99.9%) - ~89% of STIM1 is bound to Ca²⁺ in the DTS (inactive sensor); only 10% free monomer - PMCA is almost entirely unactivated (CaM not bound)

IP3 forcing function — Dolan 2014 (Supporting Material Fig S2)

Dolan 2014 uses a piecewise approach: 1. **Interpolation fit** — cubic Hermite interpolant of measured human platelet [IP3] time-course data (ADP stimulation). Data only to 30s; extrapolated beyond. 2. **Parametric form (Eq S1)** — for dose-response simulations:

$$[\text{IP3}](t) = \text{IP3_rest} \times f(t)$$

Where $f(t)$ is an exponential function with parameters a , b , c and exponent 0.6. - Second rational function term allows [IP3] to settle above baseline at long times. - Shape: rises ~5-fold within 5 s, returns to new steady level within 200 s.

For our Milestone 1 implementation, we can use a simplified version:

```
def ip3_forcing(t, ip3_rest, fold=5.5, t_peak=3.0, tau_decay=60.0):
    """IP3 forcing function (Dolan 2014 Fig S2 shape)."""
    if t <= 0:
        return ip3_rest
    rise = fold * (1 - math.exp(-t / t_peak))
    decay = math.exp(-(t - t_peak) / tau_decay) if t > t_peak else 1.0
    return ip3_rest * (1.0 + (fold - 1.0) * rise * decay)
```

This captures the essential behaviour: fast rise, slow return to slightly above baseline.

Rate constants — Ca² module (Purvis 2008 Table 1)

SERCA cycle (type 3 isoform kinetics; ref: Dode 2002)

Step	Forward	Reverse
E2 ↔ E1 (shuttling)	$k = 600 \text{ s}^{-1}$	$k = 600 \text{ s}^{-1}$
$\text{E1} + 2\text{Ca}^{2+} \text{ cyt}$	$k = 1 \times 10^4 \text{ M}^{-2} \cdot \text{s}^{-1}$	$k = 10 \text{ s}^{-1}$
$\text{E1} \cdot \text{Ca}^{2+}$		
$\text{E1} \cdot \text{Ca}^{2+} \rightarrow \text{E1P} \cdot \text{Ca}^{2+}$ (phosphorylation)	$k = 700 \text{ s}^{-1}$	$k = 5 \text{ s}^{-1}$
$\text{E1P} \cdot \text{Ca}^{2+} \leftrightarrow \text{E2P} \cdot \text{Ca}^{2+}$ (conformational)	$k = 600 \text{ s}^{-1}$	$k = 50 \text{ s}^{-1}$
$\text{E2P} \cdot \text{Ca}^{2+} \rightarrow \text{E2P} + \text{Ca}^{2+}$ (transport)	$k = 1000 \text{ s}^{-1}$	$k = 4 \times 10^4 \text{ M}^{-2} \cdot \text{s}^{-1}$
$\text{E2P} \rightarrow \text{E2}$ (dephosphorylation)	$k = 500 \text{ s}^{-1}$	$k = 1 \text{ s}^{-1}$

IP3R dynamics (type 2; ref: Sneyd & Dufour 2002)

Transition	Parameters
$\text{n} + \text{Ca}^{2+} \rightarrow \text{i1}$ (inhibition)	$k = 0.64 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $L = 0.12 \mu\text{M}$, $l = 1.7 \text{ s}^{-1}$, $k = 0.04 \text{ s}^{-1}$, $l = 0.8 \text{ s}^{-1}$
$\text{n} + \text{IP3} \rightarrow \text{o}$ (opening)	$k = 37.4 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $l = 1.7 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $k = 1.4 \text{ s}^{-1}$, $l = 2.5 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $L = 54.7 \mu\text{M}$
$\text{o} + \text{Ca}^{2+} \rightarrow \text{a}$ (activation)	$k = 4 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $L = 54.7 \mu\text{M}$, $l = 4707 \text{ s}^{-1}$, $L = 0.12 \mu\text{M}$, $k = 0.54 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $l = 11.4 \text{ s}^{-1}$
$\text{a} + \text{Ca}^{2+} \rightarrow \text{i2}$ (inhibition)	Same as $\text{n} \rightarrow \text{i1}$
$\text{o} \rightarrow \text{s}$ (shutting)	$k = 11 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $L = 54.7 \mu\text{M}$, $k = 29.8 \text{ s}^{-1}$
Channel open probability	$P_o = (0.9 \cdot \text{IP3Ra}/\text{IP3Rtotal} + 0.1 \cdot \text{IP3Ro}/\text{IP3Rtotal})$

Ca² release through IP3R

Parameter	Value	Source
Single-channel conductance ($_IP3R$)	10 pS	Purvis 2008 (ref: Zschauer 1988)
Leak conductance ($_leak$)	$0.7 \text{ pS} \cdot \text{m}^{-2}$	Purvis 2008
Nernst driving force	$_IM = RT/zF \cdot \ln([\text{Ca}^{2+}]_{\text{dts}}/[\text{Ca}^{2+}]_{\text{cyt}})$	

PMCA (type 4b; ref: Caride 2007)

Parameter	Value
KM1	0.5 mM^{-1}
KM2	1.0 mM^{-1}
kcat	8.9 s^{-1}

Parameter	Value
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Rate constants — SOCE module (Dolan 2014)

Dolan 2014 uses a Monod-Wyman-Changeux (MWC) allosteric model for STIM1-Orai1 cooperative binding and channel opening. Key parameters:

Parameter	Meaning	Value	Notes
L	Intrinsic opening equilibrium constant	Scanned	One of 3 unknowns
Ka	Association constant for STIM2 binding to Orai	Scanned	
f	Binding cooperativity factor		Negative cooperativity
n	Opening cooperativity factor		Positive cooperativity
KM	Hill coefficient for puncta entry	Scanned	One of 3 unknowns
V_PM	Half-saturation for puncta entry	Scanned	One of 3 unknowns
V_IM	Plasma membrane potential	−60 mV	Fixed
C_M	DTS inner membrane potential	−70 to −100 mV	Unmeasured; key parameter
g_SOC	Specific membrane capacitance	~2 mF/cm ²	Higher than typical (1 mF/cm ²) due to invaginations
(puncta)	SOC single-channel conductance	From MWC model	
	Max fraction of STIM2 in puncta at rest	0.2	Increased → more constitutive leak

Note: Dolan 2014 uses Monte Carlo scanning over 3 unknown parameters + 9 ICs in a 12-dimensional space. They report that <0.06% of 2.6M sampled configurations satisfy all 4 homeostatic/dynamic constraints. The SOCE module is tightly coupled to VIM — configurations responsive to IP3 require VIM > −70 mV.

Rate constants — receptor module (Purvis 2008 Table 1)

P2Y1 activation (ternary complex model; ref: Kinzer-Ursem 2007, Waldo 2004)

Reaction	Key parameters
P2Y1 + ADP → P2Y1 · ADP	$k_{-1} = 9.6 \times 10^{-4} \text{ M}^{-1} \cdot \text{s}^{-1}$, $K_d = 99 \text{ nM}$
P2Y1 → P2Y1* (activation)	$k_{-2} = 7.9 \text{ s}^{-1}$, $K_{act} = 0.0001$

Reaction	Key parameters
P2Y1* + Gq · GDP ternary complex	$k_{-1} = 0.59 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $K_q = 0.032 \mu\text{M}^{-1}$
GDP/GTP exchange	$k_{-2} = 17.8 \text{ s}^{-1}$, $k_{-3} = 1 \times 10^{-4} \text{ M}^{-1} \cdot \text{s}^{-1}$
GTP binding	$k_{-4} = 1 \times 10^{-4} \text{ M}^{-1} \cdot \text{s}^{-1}$, $k_{-5} = 8 \text{ s}^{-1}$
Gq · GTP release	$k_{-6} = 850 \text{ s}^{-1}$, $k_{-7} = 1 \times 10^{-4} \text{ M}^{-1} \cdot \text{s}^{-1}$
Gq · GTP auto-hydrolysis	$k_{\text{GTP}} = 0.013 \text{ s}^{-1}$
Cooperativity factors	$\alpha = 3.35$, $\beta = 6.62$, $\gamma = 9.85$, $\delta = 9.39$

PLC activation

Reaction	Parameters
$\text{PLC} + \text{Gq} \cdot \text{GTP} \rightleftharpoons \text{PLC}^* \cdot \text{Gq} \cdot \text{GTP}$	$k_{-8} = 0.10 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $k_{-9} = 7.78 \text{ s}^{-1}$
$\text{PLC}^* \cdot \text{Gq} \cdot \text{GTP} \rightarrow \text{PLC} + \text{Gq} \cdot \text{GDP}$ (hydrolysis)	$k_{\text{cat}} = 25 \text{ s}^{-1}$
$\text{PLC} + \text{Gq} \cdot \text{GDP} \rightleftharpoons \text{PLC} \cdot \text{Gq} \cdot \text{GDP}$	$k_{-10} = 1.61 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $k_{-11} = 0.19 \text{ s}^{-1}$
$\text{G} \cdot \text{GTP} + \text{G} \rightleftharpoons \text{Gq} \cdot \text{GTP}$	$k_{-12} = 1 \times 10^{-4} \text{ s}^{-1}$, $k_{-13} = 100 \text{ M}^{-1} \cdot \text{s}^{-1}$
$\text{G} \cdot \text{GDP} + \text{G} \rightleftharpoons \text{Gq} \cdot \text{GDP}$	$k_{-14} = 0.10 \mu\text{M}^{-1} \cdot \text{s}^{-1}$, $k_{-15} = 7.78 \text{ s}^{-1}$

PIP2 hydrolysis by PLC *

Reaction	Parameters
$\text{PLC}^* + \text{PIP2} \rightleftharpoons \text{PLC}^* \cdot \text{PIP2}$	$k_{-16} = 1 \times 10^{-4} \text{ M}^{-1} \cdot \text{s}^{-1}$, $k_{-17} = 49990 \text{ s}^{-1}$
$\text{PLC}^* \cdot \text{PIP2} \rightarrow \text{DAG} + \text{IP3}$	$k_{\text{cat}} = 9.85 \text{ s}^{-1}$

IP3 degradation

Reaction	Parameters
$\text{IP3} \rightarrow \text{IP2}$ (IP3 phosphatase)	$K_M = 24 \mu\text{M}$, $k_{\text{cat}} = 31 \text{ s}^{-1}$

Key physical constants

Constant	Value	Notes
Avogadro's number	$6.022 \times 10^{23} \text{ mol}^{-1}$	
Faraday constant	96,485 C/mol	
RT/zF at 37°C	13.4 mV ($z=2$ for Ca^{2+})	Nernst equation factor
Elementary charges per Ampere-second	6.24×10^{18}	Used by Purvis for current→flux conversion

Molecule count conversions

For 6 fL cytosol: - 100 nM = $100 \times 10^{-15} \times 6.022 \times 10^{23} \times 6 \times 10^{-1} = \sim \mathbf{361 \text{ molecules}}$ - 50 nM = $\sim \mathbf{181 \text{ molecules}}$ (IP3 resting)

For 0.12 fL DTS (2% of cytosol): - 250 μ M = $250 \times 10^{-15} \times 6.022 \times 10^{23} \times 0.12 \times 10^{-1} = \sim \mathbf{18,066 \text{ molecules}}$
Ca² in store

For 0.25 fL DTS (4% of cytosol): - 250 μ M = $\sim \mathbf{37,638 \text{ molecules}}$ Ca² in store

These numbers matter — at 100 nM cytosolic Ca² there are only ~ 360 ions. The integer count representation in the engine is at the edge of continuum validity. Stochastic effects at this level are biologically real (Purvis 2008, Sveshnikova 2015 both use stochastic simulation for this reason).

Validation targets

Observable	Expected behaviour	Source
[Ca ²]cyt resting	Stable at 40–100 nM	Purvis 2008, Dolan 2014
[Ca ²]cyt after IP3 stimulus	Peak at 200–500 nM within 15–20 s	Dolan 2014 Fig 4
[Ca ²]cyt +Caex vs –Caex	Higher and more sustained with extracellular Ca ²	Dolan 2014 Fig 4C
[Ca ²]dts after stimulus	Partial depletion, refill with SOCE	Dolan 2014 Fig 4A
Store stability in EDTA	<5 μ M [Ca ²]dts decline over 333 s without Caex	Dolan 2014
IP3 dose-response	Saturating: 5–10 $\times$ IP3 gives maximum response	Dolan 2014 Fig 5A
Ca ² spiking (single cell)	0.3–0.4 Hz, 100–300 nM amplitude	Purvis 2008, Sveshnikova 2015
ADP dose-response	Peak [Ca ²]cyt at ~ 20 s, decay ~ 1 min	Purvis 2008 Fig 5
P2Y12 knockout effect	Reduced amplitude/duration, not abolished	Design doc target
MCU knockout effect	Impaired store-operated entry, reduced function	Ghatge 2026

Sveshnikova 2025 model (confirmed from online HTML)

The online full text confirms the model from ref [37] (Sveshnikova 2015): - **29 species**, 27 differential equations, 2 fixed (Caex, thrombin) - **4 compartments**: cytosol (3 fL), DTS (1.5 fL), mitochondria (0.3 fL), PM equivalent - **Key insight from review**: PLC-Gq complex never exceeds ~ 1 molecule/cell at any time. This is the stochastic bottleneck — all downstream variability traces back here. - Calcium oscillation frequency: 0.3–0.4 Hz (at medium stimulation) - Signal shape transformation: step (thrombin) $\rightarrow$ peak (PAR1-Gq-GDP) $\rightarrow$ oscillation (cytosolic Ca²) $\rightarrow$ peak (mitochondrial Ca²) $\rightarrow$ threshold (mPTP opening) - Mitochondria act as an **integrator** of the oscillatory calcium signal - mPTP opening occurs when time-integral of [Ca²]cyt exceeds a threshold

Note: The Sveshnikova 2025 Table 2 contains all reaction equations and parameters but they are embedded as images in the HTML and not machine-readable. The parameter values are referenced as coming from refs [37] (Sveshnikova 2015), [49], [50], [51], [52], [53]. The 2015 paper used COPASI software with parameters tuned to match experimental calcium oscillation data.

Discrepancies and open questions

1. **DTS volume:** Purvis/Dolan use 1–5% of cytosol (0.06–0.3 fL). Sveshnikova 2025 uses 1.5 fL (50% of their 3 fL cytosol). This is a 5–25 $\times$ difference. May explain why Sveshnikova model sustains larger oscillations. **Decision needed.**
2. **Cytosol volume:** Purvis says 6 fL; Sveshnikova 2025 says 3 fL. Likely reflects different definitions (total cell vs cytosol only vs non-granular volume). Purvis references the measurement explicitly.
3. **SERCA copy number:** Burkhart gives $\sim 25,000$ (combined isoforms). Purvis model estimates ~ 10 for high-response configurations. Dolan 2014 Table S1 gives a **representative filtered value of 11,892** — consistent with Burkhart proteomics. The Purvis 10 value was from “high-response” configurations that likely represent unrealistic extremes.
4. **Resting $[\text{Ca}^{2+}]_{\text{dts}}$:** Range from 250 μM (Dolan, measured) to 1000 μM (Sveshnikova 2025). This significantly affects store release magnitude. Dolan’s value is a direct Fluo-5N measurement; Sveshnikova’s is a model fit.
5. **VIM (DTS membrane potential):** Unmeasured in platelets. Dolan 2014 identifies this as the most uncertain parameter and shows it must be > -70 mV for functional SOCE. Clustering analysis (Fig S4) confirms responsive configurations have **high VIM** (near -60 mV). Critical for Nernst driving force.
6. **SOCE parameters (L, Ka, n, KM):** Dolan 2014 scans these as unknowns. Their clustering analysis shows the model is **insensitive to n and KM** — SOCE flux is dominated by free STIM2 availability (regulated by Ca^{2+} dts), not puncta formation. This simplifies our implementation.
7. **IP3R gating kinetics:** Both Purvis and Sveshnikova use Sneyd & Dufour 2002 type 2 model (6 states). The Dolan Table S1 shows representative state fractions at rest. **Decision: implement the full 6-state model** — the parameters are well-characterised and oscillatory behaviour requires it.
8. **IP3R copy number:** Burkhart gives $\sim 4,850$ total (all isoforms). Dolan Table S1 representative gives 1,328. This is within the Purvis “mild-response” range (1,000–5,000). The discrepancy may reflect that not all proteomic copies are functionally active in DTS membrane.
9. **Stochastic bottleneck:** Sveshnikova 2025 identifies PLC-Gq complex (~ 1 molecule) as the source of all downstream stochasticity. This is below the continuum limit and means our deterministic ODE approach will miss cell-to-cell variability in the upstream module. Acceptable for Milestone 1 (deterministic), flag for Milestone 3.

Phasing recommendation

Milestone 1 (calcium core): species needed

Minimum set to demonstrate a Ca^{2+} transient:

CA2_CYT[c]	- cytosolic free Ca^{2+}
CA2_DTS[dts]	- DTS stored Ca^{2+}
IP3[c]	- inositol trisphosphate (IP3 as forcing function initially)
IP3R[dts]	- IP3 receptor (possibly multiple states: n, o, a, i1, i2, s)
SERCA[dts]	- SERCA pump (possibly multiple cycle states)
PMCA[m]	- plasma membrane Ca^{2+} -ATPase

Optional for Milestone 1 (add if needed for stable dynamics):

STIM1[*dt*s] - Ca^{2+} sensor
 ORAI1[m] - SOC pore
 CA2_EX[e] - extracellular Ca^{2+}

Implementation choice: IP3R complexity

Option	States	Pro	Con
Full Sneyd model	6 (n, o, a, i1, i2, s)	Matches Purvis/Dolan exactly	Complex, many parameters
Simplified Hill	1 (Po as function of [IP3], [Ca ²⁺])	Easier to test and explain	May not reproduce oscillations

Recommendation: Start with the full 6-state IP3R. The parameters are well documented, and oscillatory behaviour (which is biologically important) emerges from the biphasic Ca^{2+} dependence of IP3R — this cannot be captured by a simple Hill function.

What we still need

1. ~~Dolan 2014 Supporting Material~~ OBTAINED — Table S1 representative ICs extracted. IP3 forcing function shape documented. Clustering analysis confirms responsive configurations require high VIM and high IP3R/SERCA ratio.
2. **Sveshnikova 2025 Table S1** — supplementary not found online. However, the main paper Tables 1 and 2 (available as HTML) provide all initial conditions and reaction structure. Parameters reference Sveshnikova 2015 (ref [37]) which used COPASI. For our purposes, the Purvis/Dolan parameterisation is more thoroughly validated and better documented.
3. **Decision on DTS volume** — use 2–4% of cytosol (Purvis/Dolan consensus) unless we find a strong reason for the Sveshnikova value. Recommend starting with 4.3% (Purvis direct measurement from glucose-6-phosphatase stain).
4. **Decision on VIM** — start with –60 mV (upper bound of Dolan’s sampling range, where responsive configurations cluster per Fig S4 and clustering analysis).
5. **Calmodulin (CaM) kinetics** — Dolan Table S1 gives 20,481 copies. The CaM-PMCA activation pathway needs rate constants (not fully extracted from Table 1 image). May simplify for Milestone 1 by treating PMCA as constitutively active at low rate.